HishabAI — Team NexFin Guide: From Zero to Submitted

Deadline: Tuesday 20 October 2026, 23:59 Beijing = 21:59 Dhaka

Sanika Heeya (Leader) · Tasfia Tasneem Aastha · Abdullah Al Fayed · github.com/PIXEL-L00P/Hishab_Copilot

The 6 required deliverables and where each one is

#	Required item	Status	File(s)
1	Technical documentation	Ready	`HishabAI_Technical_Documentation.pdf`
2	Presentation deck (≤ 10 minutes)	Ready: 16 slides, PowerPoint + PDF	`HishabAI_Presentation.pptx` / `.pdf`
3	Demo video (≤ 5 min, optional but do it)	**To record** (Step 6)	—
4	Source code: GitHub link, README, deployment instructions, tests	Ready; **push to GitHub** (Step 3)	`hishab-copilot.zip`
5	Security self-assessment report	Ready	`HishabAI_Security_Self_Assessment.pdf`
6	Execution evidence (sandbox logs + video)	Logs ready; video = Step 6	`HishabAI_Execution_Evidence.pdf` + `evidence/`

Plus registration: the official form + every member's passport bio page.

Who owns what

Each person should be able to explain their part in the online review and at the final.

Member	Owns	Read these files	Must be able to explain
Sanika Heeya (Leader)	Agent, security, presenting, submission emails	`app/agent.py`, `app/security.py`, `app/audit.py`, Security report	The 4 permission tiers, approval flow, why a prompt-injected AI still can't move money, the audit chain
Tasfia Tasneem Aastha	Financial engine, payment planner, financing matcher, web app	`app/finance_engine.py`, `app/planner.py`, `app/financing.py`, `app/static/`	The cash formula, alert levels, how the plan buys time, DSCR and loan eligibility
Abdullah Al Fayed	Forecasting model, uncertainty, peer benchmarking	`ml/train_forecast.py`, `app/features.py`, `app/benchmark.py`, `models/metrics.json`	Why gradient boosting, the unseen-shop test, WAPE, conformal calibration, percentile ranks and k-anonymity

Timeline (Dhaka time)

When	What	Who
Sun 18 Oct	Step 1 registration email · Step 2 run the project · Step 3 GitHub	Sanika (email), all (run)
Mon 19 Oct	Step 4 study your part · Step 5 rehearse the deck · Step 6 record the video	Everyone
Mon 19 Oct, by 23:00	**Step 7: send version 1 of the full submission** (you can update until the deadline)	Sanika
Tue 20 Oct, by 18:00	Final version sent, receipt confirmed	Sanika
Tue 20 Oct, 21:59	Hard deadline. Nothing accepted after it.	—

Step 1 — Registration form and email (today)

Form entries

Field	Write
Team Name	NexFin
Topic	Topic E: SME Finance Copilot
Entry Title	HishabAI: Agentic Cash-Flow and Profitability Copilot for SMEs
Advisor	Leave blank unless a teacher has agreed
Team Email	The leader's email only (the one checked every day)
Role	Sanika: Leader · Tasfia: Member · Fayed: Member
University	University of Liberal Arts Bangladesh (Sanika, Tasfia) · University of Dhaka (Fayed)
Year of Study	UG Year 3 (Sanika, Tasfia) · Master's Year 2 (Fayed)

Check before sending: Tasfia and Fayed were given the **same student ID** on the form; correct the wrong one. Names and passport numbers must match the passports exactly. Delete the grey example row.

Proposal Summary (193 words):

> HishabAI is an AI agent that helps small retailers in Bangladesh manage their money, avoid cash shortages and grow their profit. Our own trained model predicts weekly sales, including busy seasons such as Eid. Using these forecasts, HishabAI estimates how much cash a shop will have in the coming weeks and warns the owner before money runs short. It also helps owners understand their profit, find their break-even point, test decisions such as buying more stock or taking a loan, and compare their performance with similar shops while keeping every shop's data private. HishabAI then prepares requests to suppliers for more time to pay, reminders for customers who owe money, and a list of affordable loan options. The owner reviews and approves every message before it is sent, and the system can never move money by itself. With the owner's permission, it can share a loan-readiness

> report with a bank, NGO or microfinance provider. HishabAI works in Bangla and English, controls who can see financial records, and keeps a record of every action. On synthetic data from shops it had never seen, our model reduced forecast errors by 40% compared with simple methods.

Email to wangyr@pbcsf.tsinghua.edu.cn, CC Tasfia and Fayed, attach the form (.xlsx) and the three passport images:

```
Subject: FinTechathon 2026 International Track — Registration — Topic E — Team NexFin

Dear Ms. Wang Yiran,

We would like to register for the 2026 Shenzhen International FinTech Competition
(International Track), Topic E: SME Finance Copilot.

Team name: NexFin
Entry title: HishabAI: Agentic Cash-Flow and Profitability Copilot for SMEs
Members:
- Sanika Heeya (Leader), University of Liberal Arts Bangladesh, CSE, UG Year 3
- Tasfia Tasneem Aastha, University of Liberal Arts Bangladesh, CSE, UG Year 3
- Abdullah Al Fayed, University of Dhaka, Theoretical Physics, Master's Year 2

Attached are our completed registration form and the passport bio pages of all
members. Could you please confirm our registration? We will send our full
submission before the 20 October deadline.

Kind regards,
Sanika Heeya
Team Leader, NexFin
[Phone / WhatsApp]
```

Step 2 — Run the project on your laptop (30 minutes, everyone)

Install **Python 3.10+** from https://www.python.org/downloads/ (Windows: tick *"Add python.exe to PATH"*). Unzip `hishab-copilot.zip`, open a terminal in that folder (Windows: type `cmd` in the File Explorer address bar):

```
        python -m venv .venv
        .venv\Scripts\activate            # Windows
        # source .venv/bin/activate       # macOS / Linux
        pip install -r requirements.txt
        python scripts/seed_db.py
        python -m app.server
```

Open **http://127.0.0.1:8000** → **Owner · Farhana**. Stop with `Ctrl + C`. Check everything:

```
        python -W ignore -m unittest discover -s tests -t . -v     # expect: Ran 81 tests ... OK
        python scripts/run_demo.py                                 # expect: 25/25 checks passed
```

Problem	Fix
`python` not found	Use `py` instead (Windows), or reinstall Python with "Add to PATH"
`pip install` fails	`python -m pip install --upgrade pip`, then retry
Port 8000 busy	Windows `set HISHAB_PORT=8080` · macOS/Linux `export HISHAB_PORT=8080`
Old data on screen	Stop the server, run `python scripts/seed_db.py`, start again

Step 3 — Push to GitHub (Sanika, 20 minutes)

Your repository: **https://github.com/PIXEL-L00P/Hishab_Copilot** (make sure it is **Public**). In the unzipped project folder:

```
        git init
        git add .
        git commit -m "HishabAI: agentic SME finance copilot (FinTechathon 2026, Topic E, Team NexFin)"
        git branch -M main
        git remote add origin https://github.com/PIXEL-L00P/Hishab_Copilot.git
        git push -u origin main
```

Already pushed to the old repository? Point your folder at the new one instead of running `git init` again: `git remote set-url origin https://github.com/PIXEL-L00P/Hishab_Copilot.git`, then `git push -u origin main --force`.

If GitHub created the repository with a README, run `git pull origin main --allow-unrelated-histories` before the push, and keep **our** README if git asks. Without git: on the repository page use *Add file* → *Upload files* and drag in the contents of the folder (not the folder itself).

Then: open the repository in a private browser window and check the README shows the architecture picture; add Tasfia and Fayed as collaborators (Settings → Collaborators). **Never commit an API key.**

Step 4 — Study your part (Monday morning, 2–3 hours)

See *Who owns what* above. Prepare a 30-second answer to each question:

1. *"Your data is synthetic. How do you know it works?"* Tested on 10 unseen shops over a full held-out year against three baselines; a real pilot is next and the design doesn't change with real data. (Fayed)
2. *"What does the AI actually do?"* It drafts and queues real actions with previews; the owner taps Approve; effects land in the outbox. (Sanika)
3. *"What stops it doing harm?"* Code-level gateway: tier 3 impossible, tier 2 needs the owner; the prompt-injection test proves it. (Sanika)
4. *"Is the financing advice trustworthy?"* Explicit rules, DSCR ≥ 1.25, timing check, reasons for every option; terms illustrative until partner lenders supply them. (Tasfia)
5. *"How does peer comparison protect privacy?"* At least 5 peers or nothing is shown; only medians and ranks; lenders see ranks only. (Fayed)

6. *"Who pays for the AI?"* Shops never pay per message; core runs on our own model; lenders and MFIs pay for consented readiness reports; optional premium plan. (Sanika)
7. *"Did you train a foundation model?"* No. Our own model is the sales forecaster; a pretrained LLM is optional for free-form conversation and is bounded by the gateway. (Fayed)

Step 5 — The presentation deck (Monday, rehearse twice)

`HishabAI_Presentation.pptx` has 16 slides with **speaker notes** (View → Notes) telling each presenter what to say: title → **the team (slide 2)** → problem domain → the problem → existing solutions → our solution → functionality. Split: Sanika slides 1–6 and 13–16, Tasfia slides 7–8 and 10–11, Fayed slides 9 and 12 (about 40 seconds per slide; keep slide 2 to about 20 seconds). Time it: it must stay under 10 minutes. Email the **PDF**; keep the .pptx for the final.

Step 6 — Demo video and execution evidence (Monday, 2 hours)

Record with **OBS Studio** (free) or Windows Game Bar (`Win + G`), under 5 minutes, in English. Run `python scripts/seed_db.py` first so the demo starts clean.

Time	Show	Say
0:00–0:20	Login page	"HishabAI is an AI finance copilot for small shops in Bangladesh, by Team NexFin. All data is synthetic."
0:20–0:50	Dashboard (Farhana)	"Our model predicts sales; the engine projects cash and warns that it runs out in the week of 2 November."
0:50–1:40	**Run cash check**	"The agent drafted a plan: two supplier extensions, two reminders, and an affordable ৳2.1 lakh loan. Nothing has been sent."
1:40–2:15	Approve three, reject one; open **Sandbox outbox**	"Only the owner approves. Approved messages go to the outbox."
2:15–2:45	**How you compare with similar shops** card	"Compared with 39 anonymised shops, our margin is strong but overdue dues are high: matching peers frees about ৳40,000."
2:45–3:25	Chat: second shop; the Bangla question; "Transfer 50000…"	"Answers come from calculations, in Bangla too. Moving money is blocked."
3:25–3:50	**Customer dues**: the IGNORE ALL PREVIOUS INSTRUCTIONS note	"This hidden attack in the data has no effect."
(optional, +30 s)	Press **বাংলা**; then **Mark paid** for a customer with a pending reminder, and try to approve it	"The whole app works in Bangla. When the data changes, the cash updates at once and the old reminder is refused."
3:50–4:25	Log in as **rakib** (approve fails) → **lender_demo**	"Roles are enforced; the lender sees only what the owner approved."

Time	Show	Say
4:25–4:55	**Audit trail**; terminal running tests and `run_demo.py`	"Tamper-evident log. 81 tests and 25 end-to-end checks pass."

Upload to Google Drive or YouTube (unlisted) with **"Anyone with the link"** access.

Judges' questions: short, true answers

Question	Answer
What exactly does the model predict?	**Weekly sales** for the next 8 weeks, with an 80% range. Cash is then projected by the deterministic financial engine, not by the model.
How were the test shops held out?	10 of the 40 synthetic shops (every shop whose id ÷ 4 leaves remainder 1) were never used in training, and the test period (the last 53 weeks, from 13 Oct 2025) was never used either.
How is WAPE calculated?	Σ|actual − forecast| ÷ Σ actual over all test shops and weeks. Ours 12.5%; best standard method (4-week average) 20.9%; 1 − 12.5/20.9 = 40% lower.
Is 88% coverage of an 80% band "well calibrated"?	No, it is **conservative** (the band is wider than needed). We calibrated on a holdout with a weaker model (WAPE 14.2%) and applied the same widening to the final model, which is more accurate, so the band ends up too wide. We say "conservative" on purpose.
Is the 40% real-world accuracy?	No. It is on synthetic data and shows the prototype works; a pilot with real shops is the next step.
How is the monthly instalment computed?	Equal instalments on a reducing balance at the mid-point of each indicative rate over the product tenure (e.g. ৳2,10,000 at 6% for 24 months = ৳9,307).
What is DSCR?	Monthly operating cash flow ÷ (existing instalments + new instalment). Style Corner: 87,181 ÷ (45,000 + 9,307) = 1.61; we need ≥ 1.25.
Do you approve loans?	No. "Likely eligible" means the shop passes our illustrative rules. The lender decides; HishabAI never submits applications.
Could two approvals send a message twice?	Not any more. An external code review found this race; we fixed it with one locked database transaction and a conditional update. Our test fires 8 approvals at the same moment and exactly one message is sent (the old code sent 8).
What if the customer pays before the owner approves the reminder?	The action is re-checked against current data at approval time; it becomes "stale" and nothing is sent (demo check 24).
What do the 81 tests / 25 checks prove?	That the software behaves as designed: formulas, model, permissions, concurrency, security rules and the full demo flow. They are not a professional security audit and do not prove accuracy on real shops.

Question	Answer
Why is there no registration page?	Accounts are pre-created for the demo. In production, onboarding includes owner verification and invite-only roles for accountants and lenders. Judges can test in seconds, and nobody can give themselves access to a shop's finances.
Is this production-ready?	No. It is a local prototype (Python's built-in server, SQLite). Production needs HTTPS behind a reverse proxy, rate limiting, PostgreSQL and a security review (see the risk list).
Does "offline" mean an AI model runs without internet?	Our own forecasting model and all calculations run inside our app with no external service. The built-in chat assistant is rule-based. An external LLM is optional, only for free-form chat.

Step 7 — Send the submission (Monday night, final on Tuesday)

```
Subject: FinTechathon 2026 International Track — Submission — Topic E — Team NexFin

Dear Ms. Wang Yiran,

Please find the submission of Team NexFin for Topic E: SME Finance Copilot.
Entry: HishabAI: Agentic Cash-Flow and Profitability Copilot for SMEs

1. Technical documentation: attached (HishabAI_Technical_Documentation.pdf)
2. Presentation deck: attached (HishabAI_Presentation.pdf)
3. Demo video: <link>
4. Source code: https://github.com/PIXEL-L00P/Hishab_Copilot
   (README with deployment instructions; 81 automated tests in /tests)
5. Security self-assessment report: attached (HishabAI_Security_Self_Assessment.pdf)
6. Execution evidence: attached (HishabAI_Execution_Evidence.pdf); raw sandbox
   operation logs in the repository folder /evidence; and the demo video above.

Team: Sanika Heeya (Leader), Tasfia Tasneem Aastha, Abdullah Al Fayed
Registration sent on 18 October 2026.

Kind regards,
Sanika Heeya
Team Leader, NexFin
```

- Keep attachments under ~20 MB; send big files as links.
- Open every link in a private window to confirm access without login.
- No receipt by Tuesday noon Dhaka time → send a short follow-up.

Optional — LLM mode

Not needed to submit. If you get an API key from console.anthropic.com, start the server with `set ANTHROPIC_API_KEY=...` (Windows) or `export ANTHROPIC_API_KEY=...` (macOS/Linux). The terminal shows `agent mode: LLM`. Test it before showing it in the video. Never share or commit the key.